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Corporate governance and voluntary disclosure

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Abstract

This paper examines the impact of ownership structure and board composition on voluntary disclosure. Ownership structure is characterized by managerial ownership, blockholder ownership and government ownership, and board composition is measured by the percentage of independent directors. Voluntary disclosure is proxied by an aggregated disclosure score of non-mandatory strategic, non-financial and financial information.

Our results show that ownership structure and board composition affect disclosure. We find that lower managerial ownership and significant government ownership are associated with increased disclosure. However, blockholder ownership is not related to disclosure. An increase in outside directors reduces corporate disclosure. We also find that larger firms and firms with lower debt had greater disclosure.
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1. Introduction

The incentive of firms to voluntarily disclose information has been of interest to both analytical and empirical researchers in accounting. Analytical

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research has examined issues such as how competition affects disclosure (e.g., Verrecchia, 1983; Darrough and Stoughton, 1990), and the use of disclosure as a signal of firm value (e.g., Hughes, 1986). Empirical research on voluntary disclosure has a much longer history, dating back to work by Cerf (1961), with a stream of subsequent studies documenting the impact of firm characteristics such as size, listing, leverage and managerial ownership on disclosure. Skinner (1994) finds that large negative earnings surprises are more often preempted by voluntary corporate disclosures. More recent research suggests that disclosure affects the cost of equity capital (Botosan, 1997) and cost of debt capital (Sengupta, 1998).

This paper examines whether corporate governance is associated with voluntary disclosure. Specifically, this paper examines the association between ownership structure, board composition and voluntary disclosure. Managerial ownership (Jensen and Meckling, 1976) and blockholder ownership (Kaplan and Minton, 1994) are two major governance mechanisms that help control agency problems. In addition, Fama (1980) argues that the board of directors is the central internal control mechanism for monitoring managers.

The framework for linking disclosure quality to corporate governance is provided in Williamson (1985). Recent empirical work on the association between disclosure and corporate governance include Forker (1992) and Chen and Jaggi (2000). Forker (1992) examines the association between corporate governance and share option disclosure. Chen and Jaggi (2000) examine the association between independent non-executive directors and comprehensiveness of information in mandatory financial disclosures. This paper is an extension of the research on corporate governance and disclosure. We extend prior work by examining corporate governance from two aspects, ownership structure and board composition, and examining disclosure in the broader context of voluntary disclosure in the financial statements.

The structure of ownership determines the level of monitoring and thereby the level of disclosure. Ownership structure is assessed by the proportion of shares held by managers and blockholders. Managerial ownership is the proportion of ordinary shares held by the CEO and executive directors. Blockholder ownership is the proportion of ordinary shares held by substantial shareholders (that is, shareholdings of 5% or more). It is expected that voluntary disclosure is negatively associated with managerial ownership and blockholder ownership. When managerial ownership is low, there is an increased need for monitoring. Similarly, there is an increased need for monitoring in diffused ownership (low blockholder ownership). In this study, we also examine the impact of government ownership on voluntary disclosure. In Singapore, significant government ownership of private sector firms is relatively common. For example, the government has between 20% and 80% ownership of around 10% of listed Singapore firms. These firms are often referred to as government-linked companies (GLCs). We expect GLCs to have

greater disclosure to mitigate the higher agency costs and weaker governance of these firms.

We expect board composition (measured by the proportion of outside directors) to be positively associated with voluntary disclosure. The role of the board of directors is to monitor management decisions. Having a higher proportion of outside non-executive directors on the board would result in better monitoring of the activities by the board and limit managerial opportunism (Fama, 1980; Fama and Jensen, 1983). Outside directors who are less aligned to management may be more inclined to encourage firms to disclose more information to outside investors. Then, it is expected that having more outside directors on the board will also result in more voluntary disclosure.

Voluntary disclosure is measured by the amount and detail of non-mandatory information that is contained in the management discussion and analysis in the annual report. We develop a disclosure scoresheet, and each sample firm's annual report is scored on the level of strategic, non-financial and financial information that is voluntarily disclosed. The disclosure score is an aggregate of the points scored by the sample firm.

The sample consists of 158 firms listed on the Stock Exchange of Singapore. In recent years, there has been an increasing call for firms in Singapore to improve on the corporate governance structure and financial disclosure. [1] The government sees corporate governance and disclosure as necessary measures to protect shareholders. [2] Shareholder protection is an issue of increasing importance in Singapore's aim to become a major financial center in Asia. Since best practices in corporate governance and greater disclosure are just being promoted, there is probably a cross-sectional variance in corporate governance

[1] Although corporate governance and disclosure standards in Singapore are often seen to be amongst the best in Asia, the widespread view, borne out by several surveys (e.g., PWC, 1997, 2000), is that they lag behind those in developed markets such as United States, United Kingdom and Australia. The Securities Association of Singapore (SIAS), an association of small minority shareholders, has indicated, "We are not happy with the current disclosure standards and would like companies to be more open" (The Straits Times, July 5, 2000, p. 39).

[2] In January 2001, the Government formed three private sector-led committees to recommend improvements in corporate governance, disclosure and accounting standards and company regulation and regulatory framework. The recommendations of the Corporate Governance Committee (CGC) and Disclosure and Accounting Standards Committee (DASC) have since been released and have been adopted by the Government. Under the CGC's recommendations, listed companies have to comply with a Code of Corporate Governance modelled after the UK Combined Code, and where they do not comply, they have to disclose this fact and give reasons for non-compliance. Under the DASC's recommendations, there will be a legal obligation for companies to make continuous disclosure of information, and accounting standards will have legal backing. Following the DASC recommendation, the Government has also announced the formation of an independent panel to approve accounting standards and to recommend improvements in corporate governance practices. This panel will be called the Council on Corporate Disclosure and Governance (CCDG).

and disclosure among firms in Singapore. Hence, Singapore firms provide an appropriate sample to examine the issue of corporate governance and disclosure at this time.

We regress the disclosure score on ownership variables and board composition after controlling for debt, firm size, growth opportunities, industry, analyst following, auditor reputation, profitability and stock performance. The findings show that lower managerial ownership is associated with increased voluntary disclosure. Blockholder ownership is not related to disclosure. [3] We also find that the level of disclosure is higher in a GLC than a non-GLC.

An increase in outside directors reduces voluntary disclosure. This result is in contrast to prior research. It appears that there is a substitute relationship between outside directors and disclosure in monitoring managers in our sample firms. We also find that larger firms have greater disclosure, while firms with lower debt disclose more information. The inverse relationship between debt and disclosure is consistent with debt being a mechanism for controlling the free cash flow problem (Jensen, 1986), reducing the need for disclosure.

Overall, the results are consistent with managerial ownership, outside directors and debt being substitutes for disclosure in corporate governance. The positive relationship between government ownership and disclosure is consistent with arguments that government ownership increases moral hazard and agency problems, and disclosure is a means of mitigating these problems.

The remainder of the paper is organized as follows. Section 2 reviews prior research on the factors that affect voluntary disclosure by firms. The hypotheses are developed in Section 3. Section 4 discusses the sample and data, and Section 5 presents the analyses and results. The paper concludes with a summary and discussion in Section 6.

2. Determinants of corporate disclosure

Prior studies find that the quality of corporate disclosure is associated with certain firm characteristics. These studies measure corporate disclosure by developing a disclosure index or score to measure voluntary disclosure in financial statements. Firm characteristics found to be associated with quality of disclosure in Singhvi and Desai (1971) include listing status and earnings margin; Chow and Wong-Boren (1987) firm size, financial leverage and proportion of assets-in-place; Meek et al. (1995) firm size, international listing status, leverage and country of incorporation. In Lang and Lundholm (1993), voluntary disclosure is measured by disclosure scores prepared by the Financial

[3] We also conduct further tests to assess whether different types of blockholders (individuals, institutions/corporations and nominees) affect disclosure. We find that the presence of individual, institutional and nominee blockholders is not associated with increased disclosure.

Analysts Federation. They find that disclosure is associated with return variability, firm size and need for financing. Skinner (1994) examines earnings-related disclosures. He finds that large negative earnings surprises are more often preempted by voluntary corporate disclosures.

Other studies have examined the relationship between ownership structure and disclosure or management forecasts of earnings. Ruland et al. (1990) hypothesize that firms that release earnings forecasts have a higher proportion of outside ownership than other firms. Their hypothesis arises from Jensen and Meckling's (1976) theory that as the manager's share ownership falls, outside shareholders will increase monitoring of manager's behavior. As the manager's share ownership falls, the manager will have increased incentives to consume perks and reduced incentives to maximize job performance. To reduce monitoring costs by outside shareholders, the manager will provide voluntary disclosure. Hence, voluntary disclosure is expected to increase with the proportion of outside ownership. Ruland et al. (1990) measure ownership structure by the percentage of voting stock owned by officers and directors. Using probit analysis, they examine whether the probability of a firm's management making an earnings forecast is explained by analysts' forecast error, absolute analysts' error, firm making a debt or equity offering and ownership structure. Their results show that as inside ownership increases, firms are less likely to provide management forecast of earnings.

El-Gazzar (1998) argues that large institutional ownership may induce a higher level of voluntary disclosure. However, based on a study of interim disclosures by Finnish firms, Schadewitz and Blevins (1998) report an inverse relationship between institutional ownership concentration and disclosure. McKinnon and Dalimunthe (1993) and Mitchell et al. (1995) both find weak support for the hypothesis that increased ownership diffusion increases the disclosure of segment information.

Forker (1992) examines the association between corporate governance and share option disclosures. Chen and Jaggi (2000) find the ratio of independent board directors is associated with mandatory disclosures. We extend the work in Chen and Jaggi (2000) by examining corporate governance in terms of ownership structure and board composition. Our measure of disclosure is also different than that in Chen and Jaggi (2000) in that we measure voluntary disclosures while they measure mandatory disclosures. We expect cross-sectional differences to be greater in voluntary disclosures than in mandatory disclosures.

3. Hypotheses

This paper examines the impact of ownership structure and board composition on voluntary disclosure by firms. Ownership structure is measured by managerial ownership, blockholder ownership, and government ownership.

Managerial ownership is the percentage of ordinary shares held by the CEO and executive directors, and includes their deemed interests. When managerial ownership is low, there is a greater agency problem. That is, the manager has greater incentives to consume perks and reduced incentives to maximize job performance. Hence, outside shareholders will increase monitoring of manager's behavior to reduce the agency problem (Jensen and Meckling, 1976). Monitoring by outside shareholders increases costs of the firm. However, monitoring by outside shareholders may be reduced if managers can provide voluntary disclosure. That is, voluntary disclosure is a substitute for monitoring. Empirical evidence in Ruland et al. (1990) shows that managerial ownership to be negatively related to disclosure. Hence it is expected that voluntary disclosure increases with decreases in managerial ownership.

> H1: Ceteris paribus, there is a negative association between managerial ownership and the level of voluntary disclosure.

Blockholder ownership is the percentage of ordinary shares held by substantial shareholders (that is, shareholdings of 5% or more). When share ownership is diffused, more monitoring is required. Empirical evidence indicates a negative relation between blockholder ownership and disclosure (McKinnon and Dalimunthe, 1993; Mitchell et al., 1995; Schadewitz and Blevins, 1998). Hence it is expected that voluntary disclosure increases with decreases in blockholder ownership.

> H2: Ceteris paribus, there is a negative association between blockholder ownership and the level of voluntary disclosure.

In Singapore, the government has vested ownership in some companies that are of strategic importance to the State. The relation between government ownership of private sector firms and disclosure has not been examined in prior research. GLCs in Singapore are incorporated under the Companies Act. They are run like other private commercial enterprises, but may have to look beyond pure profit goals and consider goals related to the interests of the nation. These goals may conflict with the commercial objectives of the enterprise (Mak and Li, 2001). That is, enhancing shareholder value may not be the primary objective of GLCs. These companies receive government funding and are also likely to have easier access to different sources of finance compared to non-GLCs. [4] Managers of GLCs are also likely to face less discipline from the

[4] According to the Business Times (4 March, 1997): "The fact that [GLCs] are part-owned (or managed) by the Singapore government enables them to raise funds much more cheaply—by up to four percentage points lower—than others". The Minister of Finance (Business Times, 23 August, 1997) noted that GLCs, being largely cash-rich, usually do not need to resort to raising bonds or bank borrowings.

market for corporate control because the government is expected to be long-term investors in these GLCs, and is unlikely to support unsolicited takeover offers for GLCs. Because of the government's vested interest in the GLCs and the conflicting objectives faced by these firms, there may be a greater need for communication with other shareholders of the firms. Hence, there may be greater disclosure in GLCs than non-GLCs.

> H3: Ceteris paribus, there is a positive association between government ownership and the level of voluntary disclosure.

We also examine corporate governance as represented by board composition, that is, the proportion of outside directors to the number of directors on the board. Outside directors who are less aligned to management may be more inclined to encourage firms to disclose more information to outside investors. That is, a positive relation between proportion of outside directors and voluntary disclosure is expected. Chen and Jaggi (2000) find empirical evidence of a positive relation between proportion of independent directors and disclosure. Hence, we hypothesize that the proportion of outside directors is positively associated with the level of voluntary disclosure.

> H4: Ceteris paribus, there is a positive association between the proportion of outside directors and the level of voluntary disclosure.

4. Sample and data

The sample for this paper is drawn from firms listed on the Stock Exchange of Singapore (SES) as at the end of 1995. [5] Our sample includes both financial and non-financial firms although financial firms are subject to different disclosure requirements in Singapore. The regulation of disclosure by financial firms are spread across a number of institutions, including SES, Monetary Authority of Singapore (MAS), Securities Industry Council, Registrar of Companies and Businesses, and the Commercial Affairs Department of the Ministry of Finance (Phan and Mak, 1999). We run a separate analysis for only non-financial firms, but the results are qualitatively similar.

Table 1 summarizes the definition and measurement of all the variables used in this paper. Most of the data are hand-collected. Data on share ownership by managers (inside directors) and blockholders are collected from the 1995 annual reports, supplemented by information reported in the Financial Highlights of Companies on the Stock Exchange of Singapore (1991–1995). Data on government ownership are collected from the Financial Highlights of Companies on the Stock Exchange of Singapore (1991–1995). Data on growth

[5] The stock exchange has now been renamed Singapore Exchange (SGX).

Table 1
Definition and measurement of variables

Variable	Definition	Measurement
Dependent variable		
DSCORE	Disclosure score	Total number of points awarded for voluntary disclosure of strategic, non-financial and financial information
Independent variables		
MOWN	Managerial ownership	The proportion of ordinary shares held by CEO and executive directors and shares in which they are deemed to have interest
BLOCK	Blockholder ownership	The proportion of ordinary shares owned by substantial shareholders (with equity of 5% or more)
GLC	Existence of government ownership	Existence of government ownership, coded as 1 if a company is government-linked (where government is a significant shareholder with 20% ownership or more) and 0 if company is not government-linked
GOVPCT	Government ownership	The proportion of ordinary shares owned by the government
OUTDIR	Board composition	Percentage of outside directors on the board
GROWTH	Availability of growth opportunities	Factor score derived from factor analysis of mean market-to-book value of equity, mean market-to-book value of assets and mean price-to-earning ratio
M/B ASSET	Market to book value of assets	Market value of firm (sum of market value of ordinary shares, preference shares, book value of long-term and short-term debt) divided by book value of total assets
M/B EQUITY	Market to book value of equity	Market value of ordinary shares divided by book value of ordinary shareholder's equity
P/E RATIO	Price-earnings ratio	Year-end price of ordinary shares divided by earnings per share
FSIZE	Firm size	Sum of market value of ordinary shares, book value of long-term and short-term debt and book value of preference shares
DEBT	Leverage ratio	Total liabilities divided by total assets
INDUSTRY	Regulated industries	Coded as 1 if firm belongs to the finance industry and 0 otherwise
AUDITOR	Auditor reputation	Coded as 1 if auditor is Big-Six firm and 0 otherwise
ANALYST	Analyst following	Number of analysts following the firm
STOCKRET	Stock return	Change in stock price over the year
ROE	Profitability	Return on shareholders' equity
ROA	Profitability	Return on assets

(market-to-book value of assets, market-to-book value of equity, price-earnings ratio), firm size and leverage are obtained from National University of

Singapore's Faculty of Business Administration (FBA) Financial Database (DBANK). [6] The disclosure index is based on the information Singapore firms provide in their annual reports to shareholders. The index is similar to that in Eng and Teo (1999) and Eng et al. (2001). [7] A scoresheet was designed for scoring firms on the amount and the level of detail of disclosures. A copy of the disclosure scoresheet is in Appendix A. Two research assistants independently read the annual reports of the sample firms, and assessed each annual report on the strategic, non-financial and financial information provided in the management discussion and analysis. Companies can score a maximum of 84 points based on the listed items in the scoresheet, but we provide room for more points for items that we may not have listed. The disclosure measure (DSCORE) produces a cross-sectional ranking of disclosure levels based on the amount of voluntary disclosure provided by firms in their annual reports. [8]

The final sample consists of 158 firms with complete data for analysis. We cover all the nine industry sectors, with 19 firms in Commerce, 12 firms in Construction, 25 firms in Finance, 8 firms in Hotels/Restaurants, 49 firms in Manufacturing, 11 firms in Multi-Industry, 16 firms in Properties, 5 firms in Services, and 13 firms in Transportation/Storage/Communications.

5. Analysis and results

We employ ordinary least squares (OLS) regression to examine the relationship between voluntary disclosure and the explanatory variables. The following model is estimated:

$$\begin{aligned}\mathrm{DSCORE} = {} & \beta_0 + \beta_1 \mathrm{MOWN} + \beta_2 \mathrm{BLOCK} + \beta_3 \mathrm{GLC} + \beta_4 \mathrm{OUTDIR} \\ & + \beta_5 \mathrm{GROWTH} + \beta_6 \mathrm{FSIZE} + \beta_7 \mathrm{DEBT} \\ & + \beta_8 \mathrm{INDUSTRY} + \beta_9 \mathrm{AUDITOR} + \beta_{10} \mathrm{ANALYST} \\ & + \beta_{11} \mathrm{STOCKRET} + \beta_{12} \mathrm{ROE} + \varepsilon \end{aligned} \quad (1)$$

where,
DSCORE = disclosure score;
MOWN = percentage of equity ownership by CEO and inside directors;

[6] This is a database containing share prices and accounting data of companies listed on the Stock Exchange of Singapore (SES). The database is constructed from data reported by the companies to the SES.

[7] Please refer to Eng et al. (2001) for details on the construction of the disclosure index.

[8] As reported in Eng et al. (2001), this disclosure score is positively correlated ($\rho = 0.5722$) with a company's achievements under the Institute of Certified Public Accountants of Singapore's *Annual Report Award*.

BLOCK = percentage of equity ownership by substantial shareholders (with equity of 5% or more);
GLC = dummy variable for government ownership, coded as 1 for GLCs and 0 otherwise;
OUTDIR = percentage of outside directors on the board;
GROWTH = factor score of growth opportunities; [9]
FSIZE = logarithm of market value of firm;
DEBT = total liabilities divided by total assets;
INDUSTRY = dummy variable for industry, coded as 1 for finance industry and 0 otherwise;
AUDITOR = dummy variable for auditor reputation, coded as 1 for Big-Six firm and 0 otherwise;
ANALYST = number of analysts following the firm;
STOCKRET = stock return measured by change in stock price over the year;
ROE = return on shareholders' equity.

The corporate governance variables are managerial ownership (MOWN), blockholder ownership (BLOCK), government-linked companies (GLC) and percentage of outside directors on the board (OUTDIR). We also include control variables that have been found in prior research to be associated with disclosure. The control variables included are growth opportunities, firm size, leverage, industry (whether the firm is a financial or non-financial firm), reputation of auditor of the firm, number of analysts following the firm, stock price performance and profitability as measured by return on equity (ROE). We also run the analyses with percentage of equity ownership by the government of Singapore substituting for GLC, [10] and return on assets (ROA) substituting for ROE. In general, the results are the same and differences are noted.

We also examine the impact of firm size, leverage and availability of growth opportunities on corporate disclosure. Larger firms are expected to disclose more information (Chow and Wong-Boren, 1987). Increased leverage is expected to reduce disclosure because leverage helps control the free cash flow problem, and the agency costs of debt are controlled through restrictive debt covenants in debt agreements rather than increased disclosure of information in annual reports (Jensen, 1986). Growth firms have greater information asymmetry and agency costs (Smith and Watts, 1992; Gaver and Gaver, 1993), and

[9] Principal component factor analysis was first applied to the three individual measures of growth opportunities—market-to-book value of assets, market-to-book value of equity, price-earnings ratio—to derive an overall factor score for the GROWTH variable. The three measures loaded on a single factor with an eigenvalue greater than one. This factor accounted for 53.13% of the variance of the individual measures. For the following analysis, we use the aggregate GROWTH measure as a proxy for growth opportunities.

[10] The average government ownership is 2%, median 0%, minimum 0%, and maximum 89%.

Table 2
Descriptive statistics for study variables ($n = 158$)

Variable	Max	Min	Mean	Median	Standard deviation
DSCORE	66	2	21.75	21.00	9.64
MOWN	0.87	0.00	0.14	0.00	0.27
BLOCK	0.98	0.20	0.62	0.63	0.15
GLC[a]	1	0	0.17	0	0.38
GOVPCT	0.89	0	0.02	0	0.10
OUTDIR	1.00	0.10	0.57	0.57	0.21
GROWTH	8.74	−0.63	0.00	−0.22	1.00
FSIZE (S$million)	27.87	17.51	20.13	19.97	1.54
DEBT	0.95	0.00	0.44	0.44	0.22
INDUSTRY	1	0	0.16	0	0.37
AUDITOR	1	0	0.77	1	0.42
ANALYST	34	0	7.65	1	10.80
STOCKRET	2.71	−0.38	0.14	0.01	0.44
ROE	30.83	−8.10	8.74	7.56	6.63
ROA	21.99	−2.91	4.59	3.53	4.03

[a] The mean value for GLC represents the percentage of firms that are GLCs.

hence growth firms are expected to disclose more information than non-growth firms.

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Table 2 presents the descriptive statistics for the variables employed in the paper. There is a wide range in the level of voluntary scores (DSCORE) in the sample. The highest disclosure score obtained is 66, and the lowest is 2. The mean disclosure score is 21.75 (median = 21.0). The distribution of managerial ownership (MOWN) is skewed; the average managerial holding is 14% but the median is less than 1%. The level of blockholder ownership (BLOCK) is high with a mean of 62%. On average, slightly more than half of the board of directors (57%) is outside directors (OUTDIR). [11] Government-linked corporations (GLC) comprise 17% of the sample.

[11] Outside director representation is higher in our sample of Singapore firms than in Chen and Jaggi's (2000) sample of Hong Kong firms. In their sample, the average representation of independent directors is 28.2%.

Table 3 presents the correlation between variables. Disclosure score (DSCORE) is positively correlated with GLC ($\rho = 0.369$), and negatively correlated with managerial ownership (MOWN) ($\rho = -0.288$) and the proportion of outside directors (OUTDIR) ($\rho = -0.157$). The univariate analysis supports HI that managerial ownership is negatively correlated with the level of voluntary disclosure. Consistent with H3, GLCs also have higher disclosure than non-GLCs. Contrary to H4, there is less voluntary disclosure when the board is made up of more outside directors. The correlation between blockholder ownership (BLOCK) and DSCORE is positive but not significant ($\rho = 0.068$). DSCORE is also correlated with the control variables firm size ($\rho = 0.345$), leverage ($\rho = -0.170$), regulated industry ($\rho = -0.196$) and return on equity ($\rho = 0.186$). The results show that managerial ownership and blockholder ownership are not correlated with one another. However, managerial ownership is negatively correlated to GLC ($\rho = -0.274$). Blockholder ownership is marginally positively related to the proportion of outside directors ($\rho = 0.138$), but managerial ownership and government ownership are not significantly related to the proportion of outside directors.

Table 4 reports the regression results. In Model 1, disclosure score is regressed on managerial ownership, blockholder ownership, GLC, proportion of outside directors and the control variables. The results show that two aspects of ownership—managerial and government—are related to disclosure. Disclosure decreases with greater managerial ownership ($\beta_1 = -7.507$, $t = 1.70$), and increases with government ownership ($\beta_3 = 6.757$, $t = 2.07$). This result of lower disclosure with increased managerial ownership is consistent with the findings in Ruland et al. (1990). They find that firms are less likely to provide management forecasts of earnings as inside ownership increases.

We also find that increased presence of outside directors is associated with reduced disclosure ($\beta_4 = -8.606$, $t = -2.08$). This result is in contrast to the finding in Chen and Jaggi (2000) who find that comprehensive financial disclosures are positively associated with the proportion of independent non-executive directors on corporate boards.

The results may be different due to the role played by the independent directors. There is greater representation of external directors on the board of Singapore firms (average = 57%) than Hong Kong firms (average = 28.2%). Outside directors may be elected by blockholders to represent their interests and may be able to acquire information directly, rather than through public disclosure. Outside directors may also act as a substitute for monitoring through public disclosure. That is, there is a negative relation between proportion of outside directors and voluntary disclosure. The results suggest that external directors in Singapore play a substitute-monitoring role to disclosure whereas in Chen and Jaggi's (2000) Hong Kong sample, they play a complementary role to disclosure.

Table 3
Pearson correlation analysis

	DSC-ORE	MOWN	BLOCK	GLC	OUT-DIR	GROW-TH	FSIZE	DEBT	INDUS-TRY	AUDI-TOR	ANA-LYST	ROE
DSCORE	1 (0.0)	−0.288 (0.0003)	0.068 (0.4023)	0.369 (0.0001)	−0.157 (0.0503)	0.109 (0.1743)	0.345 (0.0001)	−0.170 (0.0334)	−0.196 (0.0143)	0.126 (0.1181)	0.094 (0.2438)	0.186 (0.0390)
MOWN		1 (0.0)	0.003 (0.9660)	−0.274 (0.0005)	−0.108 (0.1792)	−0.090 (0.2618)	−0.182 (0.0229)	0.020 (0.8063)	0.082 (0.3070)	−0.117 (0.1486)	−0.112 (0.1640)	−0.075 (0.4091)
BLOCK			1 (0.0)	0.118 (0.1436)	0.138 (0.0850)	0.040 (0.6205)	−0.104 (0.1940)	−0.051 (0.5310)	−0.059 (0.4666)	0.059 (0.4636)	−0.023 (0.7754)	0.104 (0.2498)
GLC				1 (0.0)	−0.084 (0.2944)	−0.038 (0.6368)	0.297 (0.0002)	0.088 (0.2755)	−0.022 (0.7834)	0.126 (0.1183)	0.160 (0.0466)	0.041 (0.6541)
OUTDIR					1 (0.0)	−0.065 (0.4207)	−0.155 (0.0532)	−0.044 (0.5847)	−0.039 (0.6291)	0.007 (0.9279)	−0.001 (0.9878)	−0.007 (0.9344)
GROWTH						1 (0.0)	0.220 (0.0057)	−0.050 (0.5363)	−0.131 (0.1022)	0.074 (0.3576)	−0.031 (0.6989)	0.170 (0.0596)
FSIZE							1 (0.0)	0.001 (0.9889)	0.007 (0.9301)	0.037 (0.6513)	0.397 (0.0001)	0.170 (0.0584)
DEBT								1 (0.0)	0.465 (0.0001)	0.070 (0.3851)	−0.065 (0.4221)	0.080 (0.3753)
INDUSTRY									1 (0.0)	−0.015 (0.8541)	−0.145 (0.0714)	0.106 (0.2414)
AUDITOR										1 (0.0)	−0.028 (0.7289)	0.096 (0.2887)
ANALYST											1 (0.0)	0.149 (0.0992)
ROE												1 (0.0)

Table 4
OLS regressions of disclosure score against explanatory variables ($n = 158$)[a]

$$\begin{aligned}\text{DSCORE} = {} & \beta_0 + \beta_1\text{MOWN} + \beta_2\text{BLOCK} + \beta_3\text{GLC} + \beta_4\text{OUTDIR} + \beta_5\text{GROWTH} \\ & + \beta_6\text{FSIZE} + \beta_7\text{DEBT} + \beta_8\text{INDUSTRY} + \beta_9\text{AUDITOR} \\ & + \beta_{10}\text{ANALYST} + \beta_{11}\text{STOCKRET} + \beta_{12}\text{ROE} + \varepsilon \end{aligned} \quad (1)$$

$$\begin{aligned}\text{DSCORE} = {} & \beta_0 + \beta_1\text{MOWN} + \beta_2\text{BLOCK} + \beta_3\text{GOVPCT} + \beta_4\text{OUTDIR} \\ & + \beta_5\text{GROWTH} + \beta_6\text{FSIZE} + \beta_7\text{DEBT} + \beta_8\text{INDUSTRY} \\ & + \beta_9\text{AUDITOR} + \beta_{10}\text{ANALYST} + \beta_{11}\text{STOCKRET} + \beta_{12}\text{ROE} + \varepsilon \end{aligned} \quad (1a)$$

$$\begin{aligned}\text{DSCORE} = {} & \beta_0 + \beta_1\text{MOWN} + \beta_2\text{BLOCK} + \beta_3\text{GLC} + \beta_4\text{OUTDIR} + \beta_5\text{GROWTH} \\ & + \beta_6\text{FSIZE} + \beta_7\text{DEBT} + \beta_8\text{INDUSTRY} + \beta_9\text{AUDITOR} \\ & + \beta_{10}\text{ANALYST} + \beta_{11}\text{STOCKRET} + \beta_{12}\text{ROA} + \varepsilon \end{aligned} \quad (1b)$$

$$\begin{aligned}\text{DSCORE} = {} & \beta_0 + \beta_1\text{MOWN} + \beta_2\text{BLOCK} + \beta_3\text{GOVPCT} + \beta_4\text{OUTDIR} \\ & + \beta_5\text{GROWTH} + \beta_6\text{FSIZE} + \beta_7\text{DEBT} + \beta_8\text{INDUSTRY} \\ & + \beta_9\text{AUDITOR} + \beta_{10}\text{ANALYST} + \beta_{11}\text{STOCKRET} + \beta_{12}\text{ROA} + \varepsilon \end{aligned} \quad (1c)$$

Independent variables	Expected direction	Model 1	Model 1a	Model 1b	Model 1c
Intercept		−10.886 (−0.69)	−4.607 (−0.27)	−15.025 (−0.93)	−7.635 (−0.44)
MOWN	−	−7.507** (−1.70)	−9.122** (−2.09)	−6.155 (−1.45)	−7.725** (−1.84)
BLOCK	−	3.602 (0.60)	2.982 (0.49)	2.956 (0.51)	1.914 (0.32)
GLC	+	6.757** (2.07)		7.196** (2.22)	
GOVPCT	+		13.204 (1.52)		15.200** (1.80)
OUTDIR	+	−8.606** (−2.08)	−10.460*** (−2.45)	−6.877* (−1.71)	−8.743** (−2.11)

Table 4 (*continued*)

Independent variables	Expected direction	Model 1	Model 1a	Model 1b	Model 1c
GROWTH		−0.400 (−0.49)	−0.401 (−0.49)	−0.312 (−0.39)	−0.310 (−0.39)
FSIZE		1.883*** (2.56)	1.646** (2.14)	2.087*** (2.80)	1.810** (2.30)
DEBT		−8.530** (−1.72)	−7.374 (−1.49)	−7.718 (−1.50)	−6.584 (−1.29)
INDUSTRY		−4.816 (−1.32)	−5.490 (−1.50)	−3.057 (−0.86)	−4.096 (−1.15)
AUDITOR		1.026 (0.50)	0.889 (0.43)	0.600 (0.31)	0.344 (0.17)
ANALYST		−0.141 (−1.39)	−0.084 (−0.85)	−0.111 (−1.10)	−0.059 (−0.59)
STOCKRET		−0.466 (−0.22)	−0.851 (−0.40)	−0.135 (−0.07)	−0.557 (−0.28)
ROE		0.216 (1.51)	0.182 (1.25)		
ROA				0.033 (0.14)	−0.007 (−0.03)
F-value		3.27	3.05	3.03	2.84
Adjusted R^2		0.2061	0.1898	0.196	0.1808

[a] Coefficient for each variable is shown, with *t*-statistics in parentheses.
* Significant at 10% level (one-tailed test).
** Significant at 5% level (one-tailed test).
*** Significant at 1% level (one-tailed test).

The level of disclosure is not significantly related with blockholder ownership. We also examine whether different types of blockholders—individual/families (hereafter individual blockholders), institutions and corporations (hereafter institutional) and nominee blockholders—have different relations with corporate disclosure. In Singapore, large institutional shareholders are generally related companies, statutory boards or government. Financial institutions and mutual funds do not tend to have significant ownership in Singapore firms. It is also common practice for nominees to be significant shareholders in Singapore firms. Blockholders, whether individuals, institutions or nominees, are not associated with the level of disclosure (the results are not reported). This result of no association between blockholdings and disclosure is different than the results in McKinnon and Dalimunthe (1993), Mitchell et al. (1995) and Schadewitz and Blevins (1998). They find an inverse relation

between ownership concentration and disclosure. To the extent that managerial ownership and board composition are influencing corporate disclosure, the influence of blockholders on disclosure in our sample is not significant.

Of the control variables, larger firm size ($t = 2.56$) and lower leverage ($t = -1.72$) are related to greater disclosure. However, we find no significant relationship between disclosure and growth opportunities, industry, auditor reputation, analyst following, stock price performance and profitability.

5.1. Sensitivity analyses

In Model 1a, we replace the indicator variable GLC with percentage of equity ownership by the government (GOVPCT). The results differ from Model 1 in that GOVPCT is not significantly related to disclosure. In Model 1b, return on assets (ROA) replaces return on equity (ROE) as the measure for profitability. The results on GLC are the same as in Model 1, but managerial ownership (MOWN) is now not significant. Model 1c replaces GLC with GOVPCT and ROE with ROA. The results remain basically unchanged from Model 1. Overall, we conclude that disclosure is negatively associated with managerial ownership, proportion of outside directors, and positively associated with government ownership.

We obtain variance inflation factors (VIF) for the variables to test for multicollinearity. There is no indication that there is multicollinearity between the variables. None of the variables have a VIF value in excess of 10 which would indicate that multicollinearity may be influencing the least square estimates.

6. Summary and discussion

In this study, we examine the impact of ownership structure and board composition on corporate disclosure. We extend previous studies on the determinants of corporate disclosure in two ways. First, we examine the impact of three attributes of ownership structure on disclosure—managerial ownership, blockholder ownership and government ownership. Second, we examine the impact of board composition on corporate disclosure. Board composition is measured by the proportion of outside directors on the board. We also control for the impact of growth opportunities, firm size, debt, industry, auditor reputation, analyst following, stock price performance and profitability on corporate disclosure.

To measure disclosure, we use an aggregated disclosure score that measures voluntary disclosure of strategic, non-financial and financial information. Based on a sample of 158 Singapore listed firms, we find that lower managerial ownership and significant government ownership are associated with increased

voluntary disclosure. Total blockholder ownership is not related to disclosure. An increase in outside directors reduces voluntary disclosure. This is consistent with a substitute relationship between outside directors and disclosure in monitoring managers. The increased presence of outside directors increases the independence of the board. These directors are also able to acquire information for monitoring managers. Finally, we find that larger firms have greater disclosure, while firms with lower debt disclose more information. The inverse relationship between debt and disclosure is consistent with debt being a mechanism for controlling the free cash flow problem (Jensen, 1986), reducing the need for disclosure.

Overall, the results are consistent with managerial ownership, outside directors and debt being substitutes for disclosure in corporate governance. The positive relationship between government ownership and disclosure is consistent with arguments that government ownership increases moral hazard and agency problems, and disclosure is a means of mitigating these problems.

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Appendix A. Disclosure index

(S) Strategic information

(S-1) General corporate information:	Score		
Brief history of company	1		
Organizational structure/chart	1		
General description of business/activities	1		
Principal products	1		
Principal markets	1		
(S-2) Corporate strategy:	Score		
Statement of corporate goals or objectives	1		
Current strategy	1	2	3
Impact of strategy on current results	1	2	3
Future strategy	1	2	3
Impact of strategy on future results	1	2	3

(S-3) Management discussion and analysis:	Score		
Review of operations	1	2	3
Competitive environment	1	2	3
Significant events of the year	1	2	3
Change in sales/profits	1	2	
Change in cost of goods sold	1	2	
Change in expenses	1	2	
Change in inventory level	1	2	
Change in market share	1	2	

(S-4) Future prospects:	Score		
New developments	1	3	5
Forecast of sales/profit	1	2	
Assumptions underlying the forecast	1	2	
Order book or backlog information	1		

(S-5) Other useful strategic information:	Score		
_____	1	2	3
_____	1	2	3
_____	1	2	3

Sub-total (A)

(N) Key non-financial information

(N-l) Employee information:	Score
Number of employees	1
Compensation per employee	2
Value-added per employee	2
Productivity indicator	2

(N-2) Other useful non-financial disclosure:	Score		
_____	1	2	3
_____	1	2	3
_____	1	2	3

Sub-total (B)

(F) Financial information

(F-1) Performance indicators (not from financial statements):	Score
Historical figures for last five years or more (or as long as company's formation)	5
Turnover	1

Profit	1		
Shareholders' funds	1		
Total assets	1		
Earnings per share	1		
(F-2) Financial ratios:	Score		
Return on shareholders' funds (ROE)	1		
Return on assets	1		
Gearing ratio	1		
Liquidity ratio	1		
Other useful ratios (×1):	1		
_____	1		
_____	1		
_____	1		
(F-3) Projected information:	Score		
Cash flow forecast	3		
Capital expenditures and/or R&D expenditures forecast	3		
Earnings forecast	3		
(F-4) Foreign currency information:	Score		
Impact of foreign exchange fluctuations on current results	1	2	3
Foreign currency exposure management description	1	2	3
Major exchange rates used in the accounts	1		
(F-5) Other useful financial information:	Score		
_____	1	2	3
_____	1	2	3
_____	1	2	3
Sub-total (C)			
Total (Company DScore)			

Note: The disclosure scoresheet was previously published in Eng and Teo (1999) and Eng et al. (2001).

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